

Computer Networks (CS3001)

Sessional-II Exam

Date: Nov 5th 2025

Course Instructor(s)

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Section(s): BS(CS) A,B,C,D,E,F,G

Total Time (Hrs): 1

Total Marks: 70

Total Questions: 6

Roll No _____

Section _____

Student Signature _____

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Attempt all the questions.

INSTRUCTIONS

1. Calculator sharing is **strictly prohibited**. Each student must use their own calculator.
2. Q6 needs to be attempted on the **question paper**. e provided answer sheet for Q1-5.
3. Use a **blue or black pen only** (pencil is not acceptable for final answers).
4. Maximum allowed time is **1 hour**.

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[CLO 3: Demonstrate operations of various classical routing and switching protocol via Simulators]

Question 1

[Marks 10 +10]

Part a

Consider a data transmission scenario between a sender and a receiver using the Selective Repeat protocol.

Illustrate the exchange of packets and acknowledgments between the sender and receiver for packets numbered 0 to 10, considering the following events with window size 4:

1. Packet 2 is lost during the first transmission.
2. ACK for packet 5 is lost.
3. Timeouts occur at the sender after packets 7 and 8.
4. Packet 9 ack lost during transmission.

Show the sequence of transmissions, losses, timeouts, and acknowledgments on a timing diagram.

Part b

Draw sender finite state machines (FSMs) for a protocol RDT-3.1.

1. The channel may corrupt, lose, or delay packets and ACKs.
2. Packets can have sequence number 0,1 and 2. [0,1,2,0,1,2]
3. The sender:
 - a. Sends one packet at a time and starts a timer after each transmission.
 - b. Retransmits the packet on timeout or on receiving a duplicate ACK.
 - c. Ignores garbled ACKs.
4. No NAKs are used.

[CLO 2: Demonstrate the basics of network concepts using state-of-the-art network tools.]

Question 2

[Marks 8]

Two IoT monitoring devices, Device A and Device B, send sensor data to a cloud analytics server over TCP.

Both devices use the same retransmission timer parameters:

EstimatedRTT = 200 ms, DevRTT = 20 ms, $\alpha = 0.125$, $\beta = 0.25$.

- Device A operates over a stable wired link where RTTs stay close to 200 ms.
- Device B operates over a variable wireless link where RTTs fluctuate widely between 100 ms and 500 ms.

Tasks:

- a) Describe how DevRTT and TimeoutInterval evolve for each device. [3 marks]
- b) Without calculation, which device will retransmit more often, and why? [1 marks]
- c) Suppose Device B doubles β to 0.5. What happens to its TimeoutInterval behavior? [1 marks]
- d) At $t = 5000$ ms, Device B has EstimatedRTT = 200 ms and DevRTT = 20 ms ($\alpha = 0.125$, $\beta = 0.25$). It sends a segment S4, whose ACK arrives at $t = 5300$ ms. Immediately after that, it

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sends S5, which is lost and retransmitted. When the ACK for S5 is finally received, Device B sends S6. Calculate the TimeoutInterval for S6. [3 marks]

[CLO 2: Demonstrate the basics of network concepts using state-of-the-art network tools.]

Question 3

[Marks 7]

Host A (receiver) has a receive buffer (RcvBuffer) of 4000 bytes. Its application reads data at a constant rate of 1000 bytes per second. Host B (sender) transmits 1000-byte TCP segments at a rate of 2 segments per second.

Assume:

- The connection is established.
- All segments arrive successfully and in order.
- No retransmissions and delays are ignored.
- The sender stops sending when $rwnd = 0$.

Tasks:

- After 4 seconds, how much data is stored in Host A's buffer? [2 Marks]
- What is the advertised $rwnd$ at that moment? [1 Mark]
- After another 1 second ($t = 5$ s), what is the new $rwnd$. [1 Mark]

The first segment starts with sequence number 10 000.

- What is the sequence-number range of the segment sent at $t = 3$ s? [2 Marks]
- What ACK number will Host A send immediately after receiving it, assuming all prior data arrived correctly? [1 Mark]

[CLO 1: Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.]

Question 4

[Marks 15]

A TCP sender follows Reno-style congestion control. The round-trip time (RTT) is 100 ms, and the maximum segment size (MSS) is 1500 bytes. Losses are detected only by triple-duplicate ACKs unless otherwise stated.

- Give the average sending rate (in bytes/second) over a Reno cycle in terms of W , RTT, and MSS. [5 Marks]
- For $W = 200$ MSS, compute the average sending rate (in Mbps). [3 Marks]
- Now assume a timeout happens when $cwnd$ reaches 200 MSS. After the timeout:

$cwnd$ resets to 1 MSS,

$ssthresh$ is set to 100 MSS,

$cwnd$ grows exponentially (slow start) until it reaches 100 MSS,

then $cwnd$ increases linearly by +1 MSS per RTT (congestion avoidance) from 100 MSS up to 200 MSS before the next loss.

- How many RTTs are spent in slow start [2 Marks]?
- How many RTTs are spent in congestion avoidance [2 Marks]?

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iii) What is the average sending rate (in Mbps) over the full cycle [3 Marks]?

[CLO 1: Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.]

Question 5

[Marks 10]

Two long-lived flows share a bottleneck link with capacity $R = 80$ Mbps. Due to losses and timeouts, each flow's transport adds 20% retransmission overhead beyond its original data rate: $\lambda_{in} = 1.2$ for each flow. Assume the link is fully utilized and both flows have equal delivered totals.

- Compute each flow's delivered total rate at the bottleneck [3 Marks].
- For each flow, how much of its delivered total is retransmissions [4 Marks]?
- Interpreting the retransmissions as capacity that does not carry new original data, how many Mbps of the 80 Mbps are used by retransmissions network-wide [3 Marks]?

[CLO 1: Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.]

Question 6

[Marks 10]

Multiple choice questions

1.	C ✓	2.	B ✓	3.	B ✓	4.	A ✗	5.	B ✗
6.	C ✓	7.	B ✓	8.	B ✓	9.	B ✓	10.	B ✓

Q1. Compared to TCP + TLS over IP, QUIC's handshake design most directly aims to reduce:

- Round-trip propagation delay variance
- The number of required application-layer acknowledgements
- ☒ The number of steps before first application data
- Head-of-line blocking during congestion avoidance

Q2. Which of the following most accurately explains why QUIC eliminates "head-of-line (HOL) blocking" at the application level?

- It uses larger congestion windows to pre-empt queue buildup.
- ☒ It multiplexes independent reliable streams within one connection, each with separate loss recovery.
- It prioritizes UDP packets in router queues to avoid delay.
- It sends all streams in strict sequence using cumulative ACKs.

Q3. Which statement about QUIC's relationship to transport mechanisms is most accurate?

- QUIC discards error control to avoid user-space overhead
- ☒ QUIC reuses the conceptual principles of reliable transfer and congestion control in a new protocol above UDP

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- C. QUIC requires routers to mark packets for reliability
- D. QUIC mandates flow control only at the IP layer

Q4. When consecutive packets on a QUIC connection are dropped due to transient path congestion, how is the sending behavior adjusted?

- ☒ A. Loss is attributed to a specific stream and only that stream's rate is reduced
- B. The connection's congestion controller reacts, pacing or reducing sending across all streams
- C. Loss is ignored unless the UDP checksum fails
- D. QUIC offloads loss handling to TLS records

Q5: The following is the forwarding table of a router:

P1: 11001000 00010111 00010** **** → port 0

P2: 11001000 00010111 00011000**** → port 1

P3: 11001000 00010111 00011*** **** → port 2

otherwise → port 3

Destination D = 11001000 00010111 00010110 10100001. Which port would be selected?

- A. port 0
- ☒ B. port 1
- C. port 2
- D. port 3

Q6: The following is the forwarding table of a router:

Q1: 11001000 00010111 00010** **** → port A

Q2: 11001000 00010111 000101* **** → port B

Q3: 11001000 00010111 0001011 **** → port C

otherwise → port D

Destination D = 11001000 00010111 00010110 00000000. Which port would be selected?

- A. port A
- B. port B
- ☒ C. port C
- D. port D

Q7: Which statement is accurate about two common control-plane approaches?

- A. Distributed control computes forwarding centrally; software-defined control computes routes locally on routers.
- ☒ B. Distributed control runs routing algorithms on each router; software-defined control relies on a remote controller to compute and install forwarding state.
- C. Distributed control requires a central server; software-defined control eliminates any controller process.
- D. Both exclusively compute forwarding in input-port hardware.

Q8. Which statement correctly describes their roles?

- ☒ A. The data plane computes end-to-end routes; the control plane executes match-action at line rate.
- ☒ B. The data plane performs per-packet forwarding using existing tables; the control plane determines paths and installs forwarding state.

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- C. The control plane serializes bits on the physical medium; the data plane runs routing protocols.
D. The data plane enforces admission for guaranteed QoS in a best-effort network.

Q9. Consider two designs for installing forwarding state:

- Design X: distributed control plane, where each router computes its own forwarding state based on local and exchanged information.
- Design Y: logically centralized controller that computes and installs forwarding entries in all routers.

Which statement best characterizes the trade-off?

- A. X always converges faster than Y because it lacks a controller bottleneck
☒ B. Y can react faster to global topology changes when the controller remains reachable and has accurate global state ✓
C. X reduces control traffic compared to Y because routers do not exchange routes ✗
D. Y is strictly more stable than X regardless of network partitioning ✗

Q10. What is the stated goal for input-port processing relative to the link?

- A. Exceed link speed by buffering datagrams indefinitely.
☒ B. Complete input-port processing at line speed.
C. Delay packets to reduce burstiness.
D. Push processing into the control plane for consistency.

----- Best of luck 😊 -----